



Spatial variability in herbaceous plant phenology is mostly explained by variability in temperature but also by photoperiod and functional traits

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Abstract

Whereas temporal variability of plant phenology in response to climate change has already been well studied, the spatial variability of phenology is not well understood. Given that phenological shifts may affect biotic interactions, there is a need to investigate how the variability in environmental factors relates to the spatial variability in herbaceous species' phenology by at the same time considering their functional traits to predict their general and species-specific responses to future climate change. In this project, we analysed phenology records of 148 herbaceous species, which were observed for a single year by the PhenObs network in 15 botanical gardens. For each species, we characterised the spatial variability in six different phenological stages across gardens. We used boosted regression trees to link these variabilities in phenology to the variability in environmental parameters (temperature, latitude and local habitat conditions) as well as species traits (seed mass, vegetative height, specific leaf area and temporal niche) hypothesised to be related to phenology variability. We found that spatial variability in the phenology of herbaceous species was mainly driven by the variability in temperature but also photoperiod was an important driving factor for some phenological stages. In addition, we found that early-flowering and less competitive species characterised by small specific leaf area and vegetative height were more variable in their phenology. Our findings contribute to the field of phenology by showing that besides temperature, photoperiod and functional traits are important to be included when spatial variability of herbaceous species is investigated.

Keywords Botanical garden · Climate change · Flowering onset · Functional traits · Spatial variability · PhenObs

Introduction

During the past century, biomes around the world have experienced rapid changes in environmental conditions associated with increasing human activity (Jump and Peñuelas 2005; Steffen et al. 2015; Waters et al. 2016). Anthropogenic climate change has particularly intensified over the last several decades, resulting in higher temperatures (IPCC 2021), changes in precipitation patterns (Semmler and Jacob 2004; Dore 2005) and increased variability in environmental

conditions (Gherardi and Sala 2019; IPCC 2021). In response to these changes, in particular to the rising temperatures, plants and animals have shifted their phenology, as well as patterns of abundance and distribution (Menzel and Fabian 1999; Parmesan and Yohe 2003; Menzel et al. 2006; Büntgen et al. 1968). Besides temperature (Parmesan and Yohe 2003; Menzel et al. 2006; Büntgen et al. 2022), precipitation (Craine et al. 2012; Shen et al. 2015) and photoperiod/sunshine hours (Basler and Körner 2012; Petterle et al. 2013; Rice et al. 2021; Ren et al. 2022) are considered important drivers of plant phenology. Regarding the increasing variability in climatic conditions, up to now, research has mainly focused on the impact of temporal variations on plant

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